Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Semiconductor Process Fundamentals Masterclass

Course Code: C716

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0 · 1 October 2026

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 October 2026	Initial release of the Semiconductor Process Fundamentals Masterclass lesson plan: two-day schedule (9:30am–5:30pm, 15 instructional hours), four learning blocks and eight activities.	Dr Alfred Ang

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Course Overview

This two-day course develops the ability to identify CMOS technology process flows, analyse cross-module interactions, determine device and reliability performance, verify integration against product specifications and define accountable follow-up action. The deck organises the two topics into four visual learning blocks.

Field	Details
Course code	C716
Duration	15 hours over 2 days (9:30am–5:30pm, 7.5 instructional hours per day)
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Identify process flows and modules for specific CMOS technologies in manufacturing.
- LO2: Determine CMOS device performance and verify process integration in accordance with product specifications.

Daily Schedule

Day 1 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Introductions, learning outcomes and course overview	Facilitated orientation	Slides 1-10
9:50-10:55	65 min	Block 1A: patterning, etch and fab control	Visual explanation + evidence discussion	Slides 11-30
10:55-11:40	45 min	Activity 1: CMOS process module map	Guided integration mapping	Slides 31-33
11:40-12:40	60 min	Activity 2: pattern-transfer process window	Workbook analysis + decision review	Slides 34-37
12:40-1:10	30 min	Lunch	Break	-
1:10-2:30	80 min	Block 1B: implant, thermal, films and integrated flow	Visual explanation + cross-section review	Slides 39-58
2:30-3:25	55 min	Activity 3: implant and thermal budget	Evidence analysis	Slides 59-61
3:25-4:20	55 min	Activity 4: film deposition and CMP	Integration review	Slides 62-64
4:20-5:30	70 min	Integrated CMOS flow and Day 1 evidence review	Facilitated synthesis	Slides 65-66

Day 2 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Day 1 retrieval	Recap	-
9:50-11:20	90 min	Block 2A: integration performance, yield and reliability	Visual instruction + evidence review	Slides 67-86
11:20-12:20	60 min	Activity 5: STI integration review	Flow analysis + activity	Slides 87-89
12:20-12:50	30 min	Lunch	Break	-
12:50-2:05	75 min	Activity 6: yield and process capability	Worked example + workbook	Slides 90-93
2:05-3:20	75 min	Block 2B: SPC, packaging and future integration	Visual instruction + decision models	Slides 95-114
3:20-4:10	50 min	Activity 7: SPC CD excursion response	Evidence analysis + workbook	Slides 115-117
4:10-5:00	50 min	Activity 8: future-technology integration gate	Technology gate review	Slides 118-121
5:00-5:30	30 min	Course synthesis, review and feedback	Q&A and wrap-up	Slides 122-126

Topic-by-Topic Breakdown

Topic 1 - CMOS Process Modules and Process Flow · Slides 11-65

Learning Block 1A - Patterning, Etch and Fab Control · Slides 11-37

- The Wafer Fab as a System
- Contamination Has Multiple Pathways
- The Lithography Stack
- Mask Tone and Resist Tone
- Resist Coating and Bake
- Resolution, NA and Process Margin
- Overlay, OPC and Mask Protection
- Wet Etch versus Dry Etch
- Etch Selectivity and Profile
- Strip and Clean
- Pattern-Transfer Window
- Patterning Evidence Gate

Activity 1 - Build a CMOS Process Module Map · Slide 31

Organise the legacy module coverage into a defensible FEOL-to-BEOL flow and expose the interfaces that can propagate variation.

Activity 2 - Diagnose a Pattern-Transfer Process Window · Slide 34

Connect mask/resist/exposure decisions to etch profile and select a stable corrective action without masking the root cause.

Learning Block 1B - Implant, Thermal, Films and Integrated Flow · Slides 39-65

- Implant Dose and Energy

- Beamline Risks
- Crystal Damage and Activation
- Diffusion, Furnace and RTP
- Thermal Oxidation
- CVD, PECVD and Epitaxy
- PVD, Barrier and Seed
- Copper Dual Damascene
- CMP Mechanics and Chemistry
- Planarity Failure Modes
- Integrated FEOL and BEOL
- Cross-Module Budgeting

Activity 3 - Balance Implantation and Thermal Budget · Slide 59

Create an implant-to-anneal decision trail that balances activation, junction depth, damage recovery and channeling risk.

Activity 4 - Integrate Film Deposition and CMP · Slide 62

Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.

Topic 2 - CMOS Process Integration and Performance · Slides 67-121

Learning Block 2A - Integration Performance, Yield and Reliability · Slides 67-93

- Isolation and Well Integration
- Transistor Formation
- Yield Is Multiplicative
- Defect Density and Die Area
- Metrology by Question
- Measurement Strategy
- Electrical Performance
- Reliability Evidence
- HCI, TDDB and Electromigration
- Control Limits versus Specifications
- Capability: Cp and Cpk
- Evidence-Based Release

Activity 5 - Review an STI Integration Flow · Slide 87

Produce the process-flow and challenge analysis for a shallow trench isolation integration.

Activity 6 - Analyse Yield and Process Capability · Slide 90

Recreate and improve the legacy ProcessCapability exercise using an auditable Excel workbook with dynamic formulas.

Learning Block 2B - SPC, Packaging and Future Integration · Slides 95-121

- SPC as a Feedback System

- Select the Chart from the Data
- Variable Control Charts
- Attribute Control Charts
- Signals Beyond One Point
- Excursion Response
- Packaging Process Flow
- Wire Bond and Flip Chip
- 2.5D, 3D and Chiplets
- FinFET
- GAAFET Nanosheets
- Beyond Today's CMOS

Activity 7 - Respond to an SPC CD Excursion · Slide 115

Recreate the legacy P-chart/SPC exercises and generate the corrective-action chain for a CD excursion.

Activity 8 - Run a Future-Technology Integration Gate · Slide 118

Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.

Resources Required

- v1.0 PowerPoint/PDF learner slides
- v1.0 Learner Guide
- Eight self-contained activity folders with printable Markdown/PDF resources
- Pattern-transfer, capability and SPC Excel workbooks
- Projector, whiteboard/A3 paper, calculator and spreadsheet software
- LMS access for courseware download